

KARTHICK PATTABIRAMAN

thekarthickraman@gmail.com | +91-7358662140 | linkedin.com/in/karthickraman

SUMMARY

Senior Java Backend Engineer with 9+ years of experience in system design, microservices architecture, and building scalable distributed systems. Core contributor to Intuit's appointment scheduling platform, owning 190+ deliverables across 6 microservices that serve TurboTax and QuickBooks. I care about getting the architecture right early – whether that means rethinking how authorization calls are batched or choosing between Flink and Kafka Streams for a data pipeline. Core stack: Java 21, Spring Boot, GraphQL, Apache Kafka, Apache Flink, SQL/NoSQL databases (MySQL, DynamoDB, Elasticsearch), Redis, and AWS.

WORK EXPERIENCE

ALTIMETRIK (Client: Intuit)

Bangalore

JUL 2024 – Present

Part of the Appointments team within Intuit's Virtual Expert Platform, responsible for building and scaling the scheduling backend that serves TurboTax and QuickBooks.

- Built the foundational layers of the Schedule Management service – DAO layers on top of a managed relational (SQL) datastore, **GraphQL** APIs, shift management lifecycle (with automated slot generation and timezone handling), and the V2 Booking Management schema covering create, update, reschedule, and cancel flows. Published it to Intuit's company-wide **GraphQL federation**; this service now powers scheduling across two major product lines.
- The Calendar Segments API was doing **10+** sequential authorization calls per request. Replaced it with a single batch call and moved filtering into the policy engine, cutting response time by **~70%**.
- Wrote a per-customer rate limiter using **Redis Lua** scripts to block bot-driven appointment abuse. Uses a **sliding-window** approach with a configurable cap (**500 bookings** per **24-hour** window) and atomically adjusts counts on every booking create and cancel.
- Stood up a real-time data pipeline on **Apache Flink** that fans out appointment events to **4 downstream systems** (contact scheduler, data lake, notifications, engagement). Applied the **Template Method** design pattern so every **Kafka** producer follows the same retry and error-handling contract.
- Implemented just-in-time (**JIT**) slot calculation for all booking **CRUD** operations integrated with the workforce management system, improving data integrity and scheduling efficiency by **30%**. Also owned the full eWFM integration – slot reservation, release, rollback, **disaster recovery** config, and a cool-off period fix that resolved back-to-back scheduling conflicts.
- Owned the Booking Configuration and Booking Type APIs – in-memory caching (**Guava**, TTL-based), audit trails, and policy-driven authorization. Also added multi-participant booking support with guest validation, role-based attendee handling, and configurable max-participant limits.
- Set up the **domain event** system for appointment state changes (create, update, cancel) on **Kafka**, so downstream services react in real time. The Appointments service persists to **DynamoDB** and **Elasticsearch**; managed **schema evolution**, PII compliance, and the offline ticket booking flow for fallback scenarios.
- Led the **Java 21** and **Elasticsearch 8** migration for the Appointments service – replaced the deprecated REST high-level client with the new **Elasticsearch Java** API, updated the build toolchain, fixed test suite compatibility, and patched known CVE vulnerabilities.
- Set up end-to-end test automation using **Karate DSL** (**50+** scenarios across **6 services**) and ran **FMEA** across **7+** failure scenarios (**Redis** outages, identity service failures, tenant misconfigs) along with **Gatling** load tests for concurrent slot booking race conditions.
- Managed CI/CD across **5 environments** with **SonarQube** quality gates, and configured **Splunk** dashboards with **PagerDuty** alerting for production monitoring and on-call incident response.

Tech Stack: Java 21, Spring Boot, Maven, GraphQL (DGS), Apache Flink, Apache Kafka, Redis, DynamoDB, Elasticsearch, AWS (SNS/SQS/S3), Kubernetes, Jenkins, Karate, Gatling, OPA/Rego, Docker, Splunk

FULL CREATIVE PRIVATE LIMITED

Chennai

Software Engineer

MAY 2019 – JUL 2024

Backend engineering lead for Setmore, a SaaS scheduling platform.

- Integrated **Zoom**, **Google Meet**, and **teleport.video** into the platform and added Facebook as a booking channel, which drove a **45%** increase in Booking Page traffic from social media.
- Partnered with Facebook's developer team on the Business Extension Beta program. Post-launch, the Facebook channel contributed roughly **\$90K** in ad-driven revenue for the business.
- Built Setmore's native booking flow inside the Switchboard call-center platform, improving account manager productivity by **50%** and cutting call costs by **20%**.
- Migrated the data layer to support micro-frontend components, which sped up release cycles by **70%** and allowed the company to launch new brands (Anytime, InTheChair) on the same codebase.
- Improved Booking Page API response time by **65%** through **Redis** caching, query optimization, and **Google Datastore** index tuning.
- Led the backend API team – code reviews, mentorship for junior developers, and release ownership across the scheduling product line.

FULL CREATIVE PRIVATE LIMITED

Chennai

Jr. Software Engineer

JULY 2017 – MAY 2019

- Reduced signup API latency by **80%**, which directly sped up the new customer onboarding flow.
- Led the migration of push notifications from **GCM** to **Firebase** Cloud Messaging (**FCM**), improving delivery reliability to **98%**.

EDUCATION

A.R.M COLLEGE OF ENGINEERING AND TECHNOLOGY

Chennai

Bachelor of Engineering in Computer Science and Engineering

2012 – 2016

TECHNICAL SKILLS

Languages: Java, JavaScript

Backend & APIs: Spring Boot, Spring MVC, RESTful APIs, GraphQL (DGS), API Design, Microservices Architecture, System Design

Architecture: Distributed Systems, Event-Driven Architecture, High Availability, Performance Optimization, Design Patterns

Messaging & Streaming: Apache Kafka, Apache Flink, AWS SNS/SQS

SQL Databases: MySQL, BigQuery

NoSQL Databases: DynamoDB, MongoDB, Elasticsearch, Google Datastore

Caching: Redis, Guava

Cloud & Infrastructure: AWS (S3, Lambda, API Gateway, CodeDeploy), Google Cloud (App Engine, Cloud Run), Docker, Kubernetes

Testing & Reliability: Karate DSL, Gatling, JUnit, FMEA, Load Testing

DevOps & Observability: Jenkins, GitHub Actions, SonarQube, Splunk, PagerDuty, OPA/Rego, Git, Jira, Agile